VCU configuration record

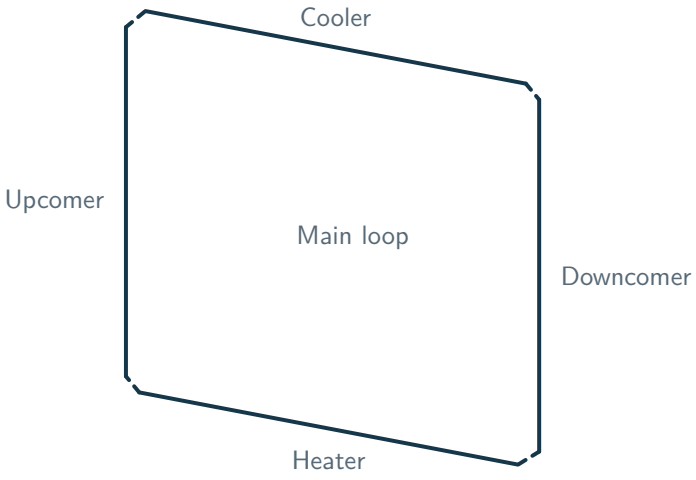
Flow-loop setup, instruments and supporting evidence

Configuration: VCU-REV9-DRAFT-002

Effective date: 2026-10-16

Status: Draft

Basis: Rev9 CAD candidate. The installed configuration has not yet been verified.



CAD overview. Dashed corners are schematic.

Read the configuration

Geometry / Instruments / Heating and cooling

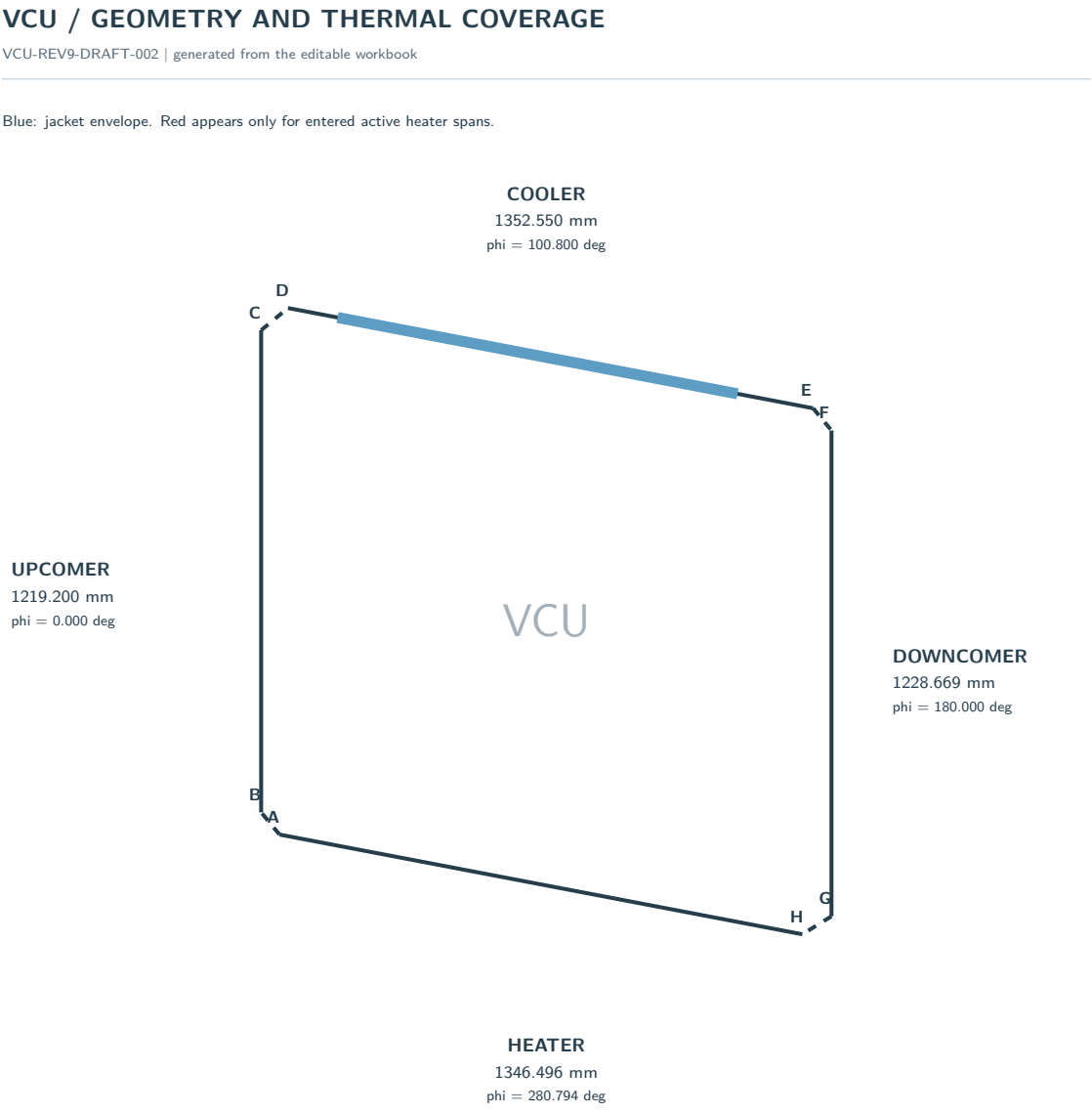
Materials and fluids / Uncertainty and limitations

Adding information? Open `VCU_inputs.xlsx` and start with one Form tab. Contributor instructions and supporting records are in the appendices.

Unknown values remain open. This draft records available evidence; it is not a released facility description.

2 Geometry

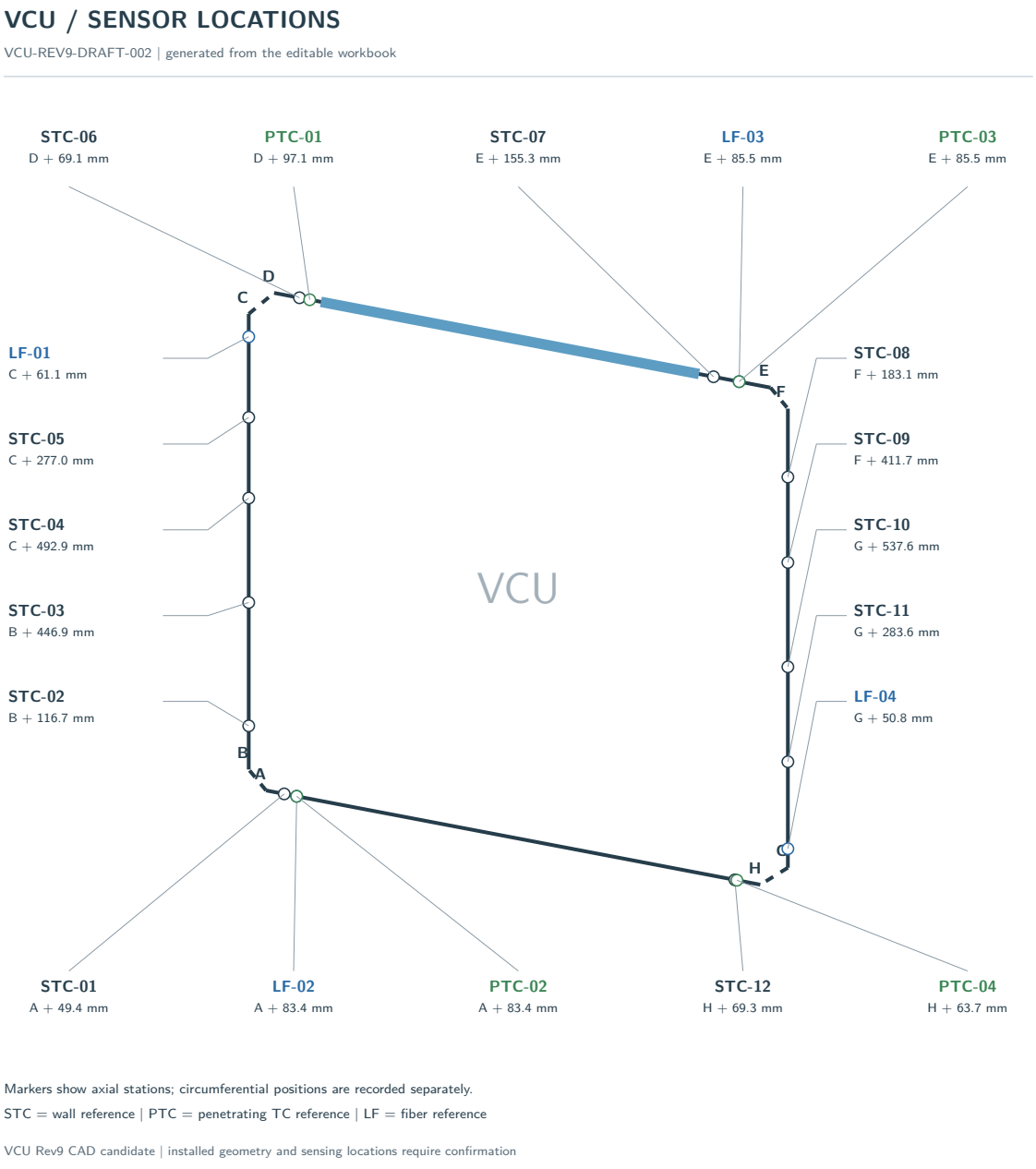
Axis-end dimensions and the recorded jacket envelope. Bore and insulation measurements remain open.



Details: coordinate conventions and CAD qualifications; dimension tables.

3 Instruments

Geometric sensor references on the main loop. Drawing IDs are not confirmed DAQ channel names.



Details: calibration and model mapping; sensor evidence form. Jacket-port sensors JTC-A/B are retained in the calibration record.

4 Heating and cooling

The mechanical jacket envelope and active thermal regions are recorded separately.

Recorded boundaries

Region	Segment / from	Start mm	End mm	Basis
Complete jacket	cooler / D	127.562	1157.849	CAD-derived
Active heater	heater / H	*Unknown*	*Unknown*	Unknown
Active cooling	cooler / D	*Unknown*	*Unknown*	Unknown

Distances are measured along the named segment from the stated endpoint. A jacket envelope does not establish active cooling; clamps do not establish energized heater coverage.

Power and control evidence

Record	Selected choice	Review
Heater supply and circuit diagram	Not selected	Needs review
Measured heater power and uncertainty	Not selected	Needs information
Heater control and monitor signals	Not selected	Needs information
Coolant routing and measurement planes	Not selected	Needs information
Coolant flow and inlet/outlet measurements	Not selected	Needs information
Cooler feedback and setpoint history	Not selected	Needs information

Supporting records: boundary and control choices; full evidence register.

5 Materials and fluids

Candidate identities are listed below. Circuit assignments and selected property models require their own evidence.

Fluids

Fluid	Product	Circuit	Basis
PROP30	Propylene glycol / water	*Unknown*	Reported
AIR	Air	*Unknown*	Reported
WATER	Water	*Unknown*	Reported

Material and property records

Record	Identity / product	Basis
PIPE	316L stainless steel	Reported
MAIN-INS	Fiberglass / reported McMaster 5556K73	Reported
HEATER-INS	Calcium aluminum silicate / reported 4057K14	Reported
PROP30	Product/formulation unresolved	Reported
AIR	Air	Unknown
WATER	Water	Unknown

Equations, units, applicability and uncertainty: full material records. To propose a property choice: material evidence form.

6 Uncertainty and limitations

The record must state how accurately the setup is known and what error is acceptable for each intended comparison.

Basis for comparison	Selected choice	Review
Dimension uncertainty and method	Not selected	Needs information
Measurement uncertainty and coverage	Not selected	Needs information
Property uncertainty and fit residuals	Not selected	Needs information
Steady-state class and acceptance limits	Not selected	Needs information
Mass/energy conservation and solution convergence	Not selected	Needs information
Validation metrics, discrepancy and applicability	Not selected	Needs information

Limits of this record

- The main-loop geometry is a CAD candidate. Installed dimensions, bore and insulation need supporting measurements.
- Sensor stations are geometric references. Channel identity, calibration, junction positions and fiber sensing extents require evidence.
- Drawn bends are schematic connectors, not qualified hydraulic paths. The heater pieces have overlapping and offset CAD geometry.
- A successful document build checks input structure and drawing closure. It does not establish experimental or modeling readiness.

Follow up: open issues; requirements and decisions; source catalog.

Appendices

Instructions, evidence and administrative detail. Return to the configuration overview at any time using the page header.

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A Contributing evidence and reviewing changes

Start with one record

You do not need to learn LaTeX to contribute. Open **VCU_inputs.xlsx**, choose a Form tab, and fill the amber response cells. The package maintainer turns the saved workbook into this document and the labeled drawings.

1. **Choose one item.** Start with the upcomer pipe, the STC-02 sensor reference, or the insulation conductivity example.
2. **Read the reference.** Gray cells show what the supplied evidence says and where to find it. They are not confirmed answers.
3. **Enter what you know.** Add units, measurement date and the point or component you checked. Leave unknown values blank; add the open question in the evidence cell.
4. **Attach proof.** Name the file and the exact photo, page, row or certificate. Choose the evidence type and add your name/date.
5. **Return it for review.** The reviewer records the decision and its scope. Form entries do not automatically replace the baseline.

A useful first return: one completed form, its evidence files, and a short list of anything you could not confirm. A partial, honest record is useful. **Keep the example attached to its item.** To work on a different segment, sensor or material, save a separate response copy. Clear the old example references or have the maintainer replace them with evidence for the new item. **Find your page:** pipe form p. 11; sensor form p. 14; property form p. 18; phase requests p. 21; complete requirements p. 31. **Need help?** The repository file **support/docs/START_HERE.md** has separate paths for the experimental team, computational team and package maintainer.

B Who fills in what?

Experimental team. Record the installed hardware, fluids, sensor/channel mapping, calibration evidence, electrical wiring, coolant routing and raw logs. Use the phase requests to see what evidence is needed next. **Computational team.** Propose model domains, boundary conditions, material correlations, uncertainty methods and validation checks. Explain the intended use and why each choice is suitable. **Package reviewer.** Resolve conflicting evidence and record the approved choice, scope, date and effective configuration. The listed teams are suggested owners; named owners and release authority still need confirmation. **Package maintainer.** Preserve the submitted evidence, update accepted input records deliberately, regenerate, and check that the document and diagrams match the saved workbook.

Two labels answer two different questions

Evidence type	What it means
CAD-derived	A result from the identified Rev9 model; installation not verified.
Installed measurement	A measured physical item with date, method and evidence.
Reported	A statement in notes or a communication; original evidence may still be needed.
Modeling assumption	A proposed idealization or convention requiring rationale.
Unknown	No supported value is available yet.

Review state is separate: Needs information, Needs review, Approved for stated use, or Not applicable with a reason. Approval must name the selected value, reviewer, date and scope. A CAD-derived value stays CAD-derived after review. This package establishes no approved physical or modeling choices.

What belongs in each package?

Configuration Control describes the physical setup and measurement chain: geometry, insulation, fluids, instruments, calibration, heating/cooling, branch states and run identity. **Modeling Control** describes how that setup is represented: 1-D and CFD/CHT domains, boundary/initial conditions, shared properties, closures, numerical checks, uncertainty and independent validation. **Shared records** connect the two: evidence, approvals, sensor-to-model mapping, measured power, property selection and validation applicability.

The STNCL documents are draft evidence sources. The requests in this package do not authorize running a test or establish that Nick Luciano has approved the draft requirements.

C Configuration record and acceptance basis

The configuration record supplies the measured inputs and uncertainties needed to compare an experiment with a model. Record what changed, which evidence applies, and what error is acceptable for the stated comparison. Do not replace an unknown with a nominal value to complete the document.

Record identity

Facility: VCU

Configuration ID: VCU-REV9-DRAFT-002

Status: Draft

Scope: Main-loop geometry and sensor-position prototype

Geometry source: VCU-R9-E01 (2026-10-08)

Prepared by: *Unknown*

Checked by: *Unknown*

Plan approval: Unconfirmed

Complete a dated package

Package item	Workbook record	Evidence to retain
Identity and changes	Start; Requirements C01–C04	Effective date, configuration ID, change history, reviewer and approved use.
Geometry and insulation	Geometry; Form pipe; C05–C16	Lengths, diameters, sensor datum, layer coverage, material certificates and measurement uncertainty.
Photographs	Requirements C17	Dated overview and detail files linked to the configuration, component and viewpoint.
Fluids and properties	Fluids; Materials; Form material; C18–C20; M12–M19	Product/composition, circuit, property source, equation, units, applicability and uncertainty.
Instrumentation and calibration	Sensors; Calibration; Form sensor; C21–C33	All instruments including fiber sensors: position, channel, certificate, correction, validity and uncertainty.
Heater and cooler	Thermal spans; C34–C41	Active boundaries, circuits, measured power/flow, calibration, feedback and setpoint history.
Run and boundary state	Model choices; C42–C46	Branches, tank/valve states, ambient conditions, operating limits, run ID and raw files.
Model and acceptable error	Model choices; Requirements M01–M32	Model revision, prediction targets, uncertainty method, conservation/convergence checks and quantitative acceptance limits.

Define what is good enough before comparison

For each prediction target, record the quantity and units, applicable run/configuration, comparison interval, error metric, quantitative acceptance limit and uncertainty coverage. Explain how measurement and model uncertainty enter the decision. Use Requirements M01, M17 and M22–M32 for the selected choices and review scope; no numerical acceptance limits are supplied by this template. **Where to enter evidence:** keep file/page/photo locators in the corresponding Requirements evidence fields. Use the selected choice, decision scope, reviewer/date and effective configuration fields for review decisions. The checklist above is a navigation aid, not a second editable record.

D Pipe and insulation segment

Example: VCU upcomer B-C. Confirm the actual section before using the reported pipe or insulation details.

Workbook tab: Form pipe. Fill columns F–I. Every unanswered line below remains **TO CONFIRM**. **Record context:** configuration / item / date / completed by: _____

Record identity C01

Reference only (CAD-derived): VCU / upcomer / B-C
Source: SEG upcomer row
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Axis-end length (mm) C06

Reference only (CAD-derived): 1219.200 mm
Source: SEG upcomer reference_length_mm
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Installed direction / tilt C07

Reference only (CAD-derived): Vertical in CAD; phi = 0 deg from +y
Source: SEG upcomer dx_mm/dy_mm
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Inside diameter (mm) C09

Reference only (Reported): CAD unknown. Notes: nominal 1.049 in (26.64 mm)
Source: DIA Main swept loop; MAT lines 23-27
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Outside diameter (mm) C10

Reference only (CAD-derived): 33.401 mm
Source: DIA Main swept loop
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Pipe and insulation segment — continued

Record context: configuration / item / date: _____

Pipe grade / specification C11

Reference only (Reported): 316L, nominal 1-inch schedule 40 reported
Source: MAT lines 23-27
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

How measured / uncertainty C12

Reference only (Unknown): No installed measurement uncertainty supplied
Source: P0 geometry / uncertainty request
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Insulation product and layer order C13

Reference only (Reported): Most piping: fiberglass, reported 5556K73
Source: MAT lines 29-35
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Thickness of each layer (mm) C14

Reference only (Reported): About 2 in for most main piping; this segment unverified
Source: MAT lines 29-35
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Coverage, gaps and condition C15

Reference only (Unknown): Unknown for this exact segment
Source: P0 geometry/configuration request
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Pipe and insulation segment — continued

Record context: configuration / item / date: _____

Contacts, supports and outer covering C16

Reference only (Modeling assumption): Firm contact is an assumption, not an installed observation
Source: HEAT VCU Insulation lines 7-10
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Photo, measurement date and measurer C17

Reference only (Unknown): Attach a photo with B and C marked
Source: Proposed survey record
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Review decision and approved scope C03

Reference only (Unknown): Needs review; include configuration ID and effective date
Source: No approval supplied
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

E Sensor, channel and calibration

Example: drawing reference STC-02. Identify the actual hardware and channel independently; a drawing label is not a DAQ mapping.

Workbook tab: Form sensor. Fill columns F–I. Every unanswered line below remains **TO CONFIRM**. **Record context:** configuration / item / date / completed by: _____

Drawing reference / installed hardware ID C21

Reference only (CAD-derived): STC-02; CAD surface-thermocouple reference
Source: SNS sensor_id=STC-02
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Manufacturer, model and serial C22

Reference only (Unknown): Unknown
Source: P0 instrumentation request
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Exact DAQ channel / units C23

Reference only (Unknown): Unknown; do not enter STC-02 as channel without checking
Source: SNS channel_label_in_CAD blank
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Segment, endpoint and distance (mm) C24

Reference only (CAD-derived): Upcomer; B + 116.686 mm
Source: SNS nearest_endpoint / nearest_distance_mm
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

What point did you measure? C26

Reference only (CAD-derived): Selected nearest pipe-surface point; not a verified junction
Source: SNS reference_basis
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Sensor, channel and calibration — continued

Record context: configuration / item / date: _____

Circumferential angle (deg) C25

Reference only (Modeling assumption): Unknown. Vertical-pipe zero is proposed drawing +x
Source: DESIGN coordinate convention; SNS angle not supplied

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Radius / insertion / active extent (mm) C26

Reference only (Unknown): Unknown for actual sensing point
Source: SNS physical_junction_is_defined=False

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Attachment / insulation position / response C27

Reference only (Unknown): Unknown installed coupling
Source: P0 instrumentation request

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Last calibration date and certificate C28

Reference only (Unknown): Unknown; prior bath claim is not an instrument certificate
Source: P0 channel table / calibration request

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Calibration range, correction and residual C29

Reference only (Unknown): Unknown
Source: P0 instrumentation request

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Sensor, channel and calibration — continued

Record context: configuration / item / date: _____

Uncertainty, units and coverage factor C30

Reference only (Unknown): Unknown
Source: P0; P1 uncertainty method

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Sample rate, filtering and time response C31

Reference only (Unknown): Unknown
Source: P0 instrumentation/logging

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Clock / synchronized event evidence C32

Reference only (Unknown): Unknown
Source: P0 logging and synchronization

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Mandatory, alternate or diagnostic use C33

Reference only (Unknown): Not assigned; requires intended-use review
Source: P0 acceptance-channel matrix

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

1-D cell and CFD region / observed quantity M04

Reference only (Unknown): Unmapped
Source: P0 geometry/model map

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Sensor, channel and calibration — continued

Record context: configuration / item / date: _____

Next calibration / recheck trigger C28

Reference only (Unknown): Unknown; include validity at the date of the experiment
Source: Proposed calibration-maintenance field

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Review decision and approved scope C03

Reference only (Unknown): Needs review; include configuration ID and effective date
Source: No approval supplied

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

F Material-property selection appendix

Example: thermal conductivity of reported main-pipe fiberglass insulation. The equation is a candidate from notes, not a selected correlation.

Workbook tab: Form material. Fill columns F–I. Every unanswered line below remains **TO CONFIRM**.
Record context: configuration / item / date / completed by: _____

Material and component M12

Reference only (Reported): Main-pipe fiberglass insulation; product 5556K73 reported
Source: MAT VCU Insulation lines 29-35
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Property to select M13

Reference only (Reported): Thermal conductivity k(T)
Source: MAT lines 38-40
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Candidate equation / table M13

Reference only (Reported): k = 5.84142e-7*T^2 - 2.89613e-4*T + 6.78938e-2
Source: MAT lines 38-40; candidate only
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Input variable and output units M14

Reference only (Reported): Output W/(m K) reported; confirm T convention against original fit
Source: MAT lines 38-40
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Source-supported domain M15

Reference only (Reported): Notes report 297-533 K; original data/domain unverified
Source: MAT line 40
Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Material-property selection appendix — continued

Record context: configuration / item / date: _____

Actual use range / margin to source domain C45

Reference only (Unknown): VCU experiment and insulation-temperature ranges unknown
Source: No operating range supplied

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Original source / edition / table or equation M13

Reference only (Reported): Product link in notes; original fit data not yet retained here
Source: MAT line 30; line 38 fit

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Evidence this product is installed M12

Reference only (Unknown): Unknown for the selected segment and layer
Source: Purchase record / survey needed

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Source uncertainty versus fit residual M17

Reference only (Unknown): Both unknown; do not treat fit precision as uncertainty
Source: P0 / P1 property uncertainty request

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Interpolation and outside-domain policy M16

Reference only (Unknown): Not selected
Source: Proposed implementation record

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Material-property selection appendix — continued

Record context: configuration / item / date: _____

Why this source; alternatives considered M18

Reference only (Unknown): Not selected; compare product data and conditions
Source: User requested rationale

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Check values, implementation and version M19

Reference only (Unknown): Not implemented or independently checked
Source: Proposed property tests

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Approved use, reviewer and date C03

Reference only (Unknown): No approved property choice established
Source: Review required

Your entry: TO CONFIRM _____
Evidence: file + page/photo/row: _____
Entry type / reviewer and date: Unknown / Pending

Before selection: verify the original data, temperature convention, units and property-specific domain. The reported 297–533 K interval is a fit domain, not a VCU operating limit. Repeat this form for every required property and material.

G What the draft phases ask the team to return

These are concrete information requests extracted from draft STNCL Phases 0–4. Values and test conditions in those plans remain draft. The electrical-heater cautions below must be resolved for VCU. **How to use:** pick a request, gather the listed files, and return them with the configuration/run ID. The computational team supplies the analysis where indicated. Mark completion only after the review partner has assessed the evidence.

P0-01 | Identify the physical setup

Please do: Photograph the whole VCU loop; name the revision, date and changed components.
Return: Configuration ID, change list and labeled photos
VCU adaptation / caution: No electrical-specific adaptation identified.
Review partner: Facility lead / package reviewer
Traceability: C01-C17; P0 Section 3 Method: configuration and geometry
Done when: Evidence supplied and reviewed; open limitations recorded.

P0-02 | Trace every channel

Please do: Match each installed sensor to its cable/DAQ channel and actual sensing point.
Return: Channel crosswalk, position survey, calibration certificates and uncertainties
VCU adaptation / caution: PTC-X omitted from the active register by owner request, 2026-10-16. Verify remaining installed sensor identities; do not assume CAD IDs are DAQ channels or TC/Luna co-location.
Review partner: Experimental + computational
Traceability: C21-C33; M04; P0 Method: instrumentation
Done when: Evidence supplied and reviewed; open limitations recorded.

P0-03 | Document heater and cooler measurement boundaries

Please do: Supply the actual circuit/plumbing and feedback sketches, meter positions and logged signals.
Return: Electrical wiring/power record; coolant routing, flow and temperature planes
VCU adaptation / caution: Replace heater secondary-fluid duty with an approved electrical control-volume balance.
Review partner: Experimental + computational
Traceability: C34-C41; M08-M09; P0 Section 2 duties; Section 3 secondary-side boundaries
Done when: Evidence supplied and reviewed; open limitations recorded.

P0-04 | Document fluid, branches and operating limits

Please do: Identify each circuit fluid, batch, fill level, valves, vent/pressure and approved limits.
Return: Fluid/inventory record, branch state sheet and limit references
VCU adaptation / caution: Draft Prop-30 limits do not automatically apply to air or water experiments.
Review partner: Facility lead / package reviewer
Traceability: C18-C20; C42-C45; M07; P0 Method: facility/fluid configuration and operating limits
Done when: Evidence supplied and reviewed; open limitations recorded.

P0-05 | Demonstrate logging and synchronization

Please do: Provide a checkout record showing required signals and a common timed event.
Return: Raw checkout file, channel units/rates, clock offsets and event log
VCU adaptation / caution: Confirm two room T/RH stations and exposed cold-surface measurements; dew-point rule remains draft.
Review partner: Experimental + computational
Traceability: C31-C33; C44; C46; P0 Method: environment, synchronization and coordinated checkout
Done when: Evidence supplied and reviewed; open limitations recorded.

P1-01 | Supply a qualified ambient baseline

Please do: Under the approved procedure, record the passive baseline and room/control history.
Return: At least 6 h after ambient equilibrium plus a relevant HVAC cycle, as requested by draft
VCU adaptation / caution: Verify electrical heater input and residual cooler duty are inactive; command OFF alone is insufficient.
Review partner: Experimental + computational
Traceability: C44; C46; M22-M24; P1 Section 3 Method, Record the baseline
Done when: Evidence supplied and reviewed; open limitations recorded.

P1-02 | Explain channel quality and substitutions

Please do: Identify bad channels; provide real TC/Luna pairs and note spatial offsets.
Return: Quality/exclusion list, pairing map, noise/drift/covariance summary
VCU adaptation / caution: P0 says not all locations are co-located; P1 wording needs a usable-pair rule.
Review partner: Experimental + computational
Traceability: C23-C33; M21; M24; P1 Section 3 Required Outputs
Done when: Evidence supplied and reviewed; open limitations recorded.

P1-03 | Agree on measurable acceptance criteria

Please do: Review whether proposed limits can be resolved by the baseline measurements.
Return: Approved channel roles, block/window rules and resolution evidence
VCU adaptation / caution: Illustrative targets are not approved VCU limits; improve measurement/window before relaxing intent.
Review partner: Computational team
Traceability: C33; M22-M24; P1 Section 2 and Section 3 Required Outputs
Done when: Evidence supplied and reviewed; open limitations recorded.

P2-01 | Agree on the conditioning procedure

Please do: Confirm how VCU is filled, circulated, isolated and judged near-isothermal.
Return: Reviewed procedure, fill/inventory/valve history and initial temperature map
VCU adaptation / caution: Resolve continuous in-loop recirculation versus tank-uniformity-before-each-fill wording.
Review partner: Experimental + computational
Traceability: C20; C43; M20; P2 Section 3 Method and Required Outputs step 1
Done when: Evidence supplied and reviewed; open limitations recorded.

P2-02 | Preserve warm and cold relaxation records

Please do: Provide full conditioning, shutdown and relaxation histories with room/event data.
Return: Two warm and two cold runs requested by draft; calibration/withheld roles assigned in advance
VCU adaptation / caution: Draft target offsets/durations are provisional; establish VCU operating and condensation limits first.
Review partner: Experimental + computational
Traceability: C46; M20; M25; P2 Section 3 Run IDs and Method
Done when: Evidence supplied and reviewed; open limitations recorded.

P2-03 | Demonstrate the passive boundary

Please do: Show actual electrical input and residual cooler/attached-hardware heat exchange after shutdown.
Return: Residual-duty evidence, isolation timing and initial-state uncertainty
VCU adaptation / caution: Include heater stored heat; zero electrical input does not prove zero thermal release.
Review partner: Experimental + computational
Traceability: C36; C40; M06; M08; M20; P2 Method initial condition and Required Outputs qualification
Done when: Evidence supplied and reviewed; open limitations recorded.

P2-04 | Review the fitted model before later tests

Please do: Confirm the sensor-to-zone map and explain structured residuals with experimental context.
Return: Frozen model/property/mapping versions, withheld comparisons, parameter uncertainty and settling bound
VCU adaptation / caution: Computational team owns fits; experimental team supplies physical-state evidence.
Review partner: Computational team
Traceability: M21; M25-M27; P2 Section 3 Required Outputs / Disposition
Done when: Evidence supplied and reviewed; open limitations recorded.

P3-01 | Provide low-duty accepted-state records

Please do: Return heater-only and cooler-only histories, inactive-system evidence and accepted intervals.
Return: HTR-LOW-01 / CLR-LOW-01 records, ambient exchange and uncertainty inputs
VCU adaptation / caution: Adapt whole-loop energy accounting to measured electrical input and explicit losses/storage.
Review partner: Experimental + computational
Traceability: C36; C40; C44; M22-M24; P3 Section 3 Method / Required Outputs
Done when: Evidence supplied and reviewed; open limitations recorded.

P3-02 | Keep blind comparisons independent

Please do: Review discrepancies before any fitted parameter is changed.
Return: Dated blind comparison, residuals, condensation class and disposition
VCU adaptation / caution: A run reassigned to calibration cannot validate the revised model; obtain new independent cases.
Review partner: Computational team
Traceability: M25; M31; P3 Section 2 Blind validation; Section 3 Disposition
Done when: Evidence supplied and reviewed; open limitations recorded.

P4-01 | Agree on attainable target bands

Please do: Confirm instrumentation and controls can resolve the proposed mean-temperature and duty targets.
Return: Approved target coordinates/bands and actual achieved conditions
VCU adaptation / caution: Replace heater-duty coordinate with agreed electrical/net-thermal definition consistently.
Review partner: Experimental + computational
Traceability: C36; C41; M22-M23; P4 Section 3 Method: select matrix coordinates
Done when: Evidence supplied and reviewed; open limitations recorded.

P4-02 | Return complete matrix and transition histories

Please do: Preserve OP-A/B/C/D/E and B-R records, approach paths, events and failed attempts.

Return: Five unique states plus repeat; accepted spans, reporting intervals and transition times

VCU adaptation / caution: State constancy and target attainment are separate checks; no retroactive threshold relaxation.

Review partner: Experimental + computational

Traceability: C46; M23; M32; P4 Section 3 Run IDs / Required Outputs

Done when: Evidence supplied and reviewed; open limitations recorded.

P4-03 | Qualify inference and applicability

Please do: Provide usable exchanger-temperature differences and uncertainty inputs; identify a withheld applicability state.

Return: Flow/Reynolds conditioning evidence, independent estimates and applicability comparison

VCU adaptation / caution: Heater-side inference needs the electrical adaptation; no quantitative inference from unresolved heat balance.

Review partner: Computational team

Traceability: M15; M25; M30-M31; P4 Section 2 flow inference; Section 3 Required Outputs

Done when: Evidence supplied and reviewed; open limitations recorded.

H Issues to resolve together

All issues below are open. The suggested owners help route a question; they do not imply an assigned or approved responsibility.

I01 | Hardware / plan mismatch

Finding: Plans use heater secondary enthalpy flow; VCU notes report electrical mat/band heaters.
Next decision: Define electrical measurement boundary, real power, heater/structure storage and external losses; revise Phase 0-4 references.
Suggested owner: Experimental + computational
Source / locator: P0 Section 2 duties; P3 energy balance versus HEAT VCU Heater
Affected fields: C34-C37; M08; M30

I02 | Internal procedure inconsistency

Finding: In-loop recirculation is specified, but qualification still refers to tank uniformity before each fill.
Next decision: Choose the actual conditioning method and align initial-condition, timing and acceptance wording.
Suggested owner: Plan author + experimental lead
Source / locator: P2 Section 3 Method vs Required Outputs step 1
Affected fields: C20; M20

I03 | Conditional measurement assumption

Finding: P0 states not all sensors are co-located; P1 requests each TC compared with its co-located Luna measurement.
Next decision: Identify actual pairs and offsets; define how non-coincident pairs may be used.
Suggested owner: Instrumentation + computational
Source / locator: P0 apparatus/channel table vs P1 Required Outputs TC-Luna comparison
Affected fields: C24-C27; C33; M21

I04 | Conflicting sensor evidence

Finding: Historical CAD extraction includes PTC-X; the supplied note calls PTCX a fluke. PTC-X was removed from the active sensor/calibration records by owner request, 2026-10-16. Installed hardware identity is unverified.
Next decision: Retain the original CAD/source evidence. Confirm whether the excluded reference corresponds to installed hardware before using it in any run or model mapping.
Suggested owner: Experimental instrumentation
Source / locator: R9 sheets 5-6; SNOTE VCU Rev9 PTCX note
Affected fields: C21; C23

I05 | Unqualified geometry interpretation

Finding: Heater pieces overlap/offset; lower-right bend is reconstructed; clamp spans are mechanical only.
Next decision: Survey geometry; document any model healing and active heater boundaries separately.
Suggested owner: Experimental geometry + computational
Source / locator: R9 sheets 2-4 and SEG
Affected fields: C06-C08; C34; M03

I06 | Cross-facility / revision contamination

Finding: Some VCU note links target ACU or Rev8; supplied VCU overview illustrations identify Rev8.

Next decision: Keep references for context only; use Rev9 rows as CAD candidate and new installed evidence for verification.

Suggested owner: Package maintainer

Source / locator: SNOTE VCU sections; supplied VCU illustration captions

Affected fields: C04; C23

I07 | Applicability gap

Finding: Draft campaign is Prop-30-specific; user also lists air/water VCU experiments.

Next decision: Keep plan thresholds and fluid correlations scoped to the documented fluid/circuit and requalify other uses.

Suggested owner: Facility lead + computational

Source / locator: P0 Purpose/Required scope; USER fluid clarification

Affected fields: C18-C19; C45; M15

I08 | Approval status missing

Finding: Documents are marked draft; no Nick-approved requirement status is established here.

Next decision: Confirm approved revision and record which requirements remain proposed.

Suggested owner: Facility lead / plan author

Source / locator: P0-P4 title/status and approval records

Affected fields: C03; M22

I09 | Property-source qualification gap

Finding: Notes contain candidate fits/mixed-source choices; original validity and uncertainty are not yet verified.

Next decision: Check original sources and installed identity before approving equations; keep source domain separate from operating limits.

Suggested owner: Computational materials + experimental

Source / locator: MAT lines 35-40; STEEL lines 3-21; PG figures

Affected fields: M12-M19

I Rev9 geometry and drawing conventions

Only the qualified Rev9 structured geometry seeds the CAD candidate. Original source rows are retained in support/reference/rev9_extraction.json. Installed measurements, reported nominal sizes and selected model geometry remain separate records.

- Main-loop bore remains unknown in the CAD baseline. Local spool bores cannot supply the missing loop-wide value.
- Heater pieces overlap by about 7.202 mm and their axes differ by about 0.152 mm. The displayed heater length is a projected span, not an exact continuous hydraulic length.
- The G–H bend is reconstructed from spline geometry with qualified fit residuals. Dashed overview connectors are schematic; they do not define hydraulic arc lengths.
- Jacket shell and full envelope differ. Neither establishes the active cooling span; clamps do not establish active electrical heating spans.
- Sensor markers show geometric references. Physical junctions, fiber extents, channels and circumferential positions require evidence.

Measured distances: choose the segment and the labeled endpoint. Measure along its axis. Record which physical point you measured. For a bent section, ask the maintainer for a qualified path convention before entering an arc distance. **Circumferential position:** this is the location around the pipe, separate from the leg inclination. For a vertical leg, "top of the pipe is ambiguous; the proposed zero is drawing +x. The convention needs team agreement.

Coordinate definitions for the maintainer

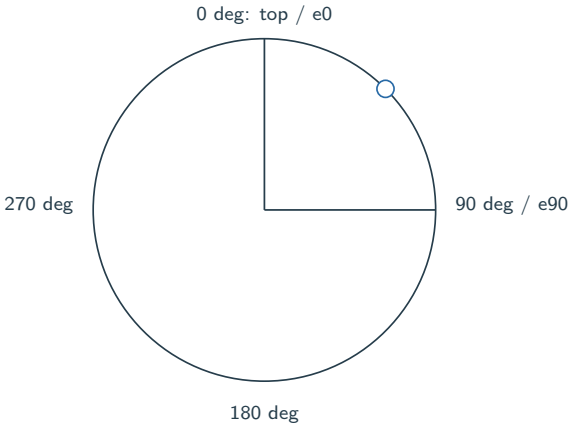
Origin B; x rightward, y upward, n out of the drawing. Straight-leg direction ϕ is clockwise from upward vertical. Thus $\Delta x = L\sin\phi$ and $\Delta y = L\cos\phi$. Circumferential angle α is a separate input. For distance d from the start, $s = d$; from the end, $s = L - d$. Then $\mathbf{p}_a = \mathbf{p}_0 + s\mathbf{t}$. Define $\mathbf{e}_0$ as projected upward vertical perpendicular to $\mathbf{t}$, normalized; for vertical pipes use drawing $+x$. Let $\mathbf{e}_{90} = \mathbf{t} \times \mathbf{e}_0$. The point is $\mathbf{p} = \mathbf{p}_a + r(\cos\alpha\,\mathbf{e}_0 + \sin\alpha\,\mathbf{e}_{90})$. Both radius and angle are required for a full point. A blank is unknown, not zero. Printed precision is not measurement uncertainty.

Current generator qualifications

- Active heater: boundaries unknown; not drawn.
- Active cooling: boundaries unknown; not drawn.
- CAD references are not verified sensing junctions or acquisition-channel names.
- Bends are schematic connectors. No continuous hydraulic length is inferred.
- JTC-A/B, branch pipes, tank and coolant plumbing are retained as outstanding scope, not drawn.
- Model input draft only: missing bore, property selections, calibrations and approved boundary conditions.

SENSOR POSITION / ANGLE CONVENTION

Circumferential angle is separate from segment inclination.



HOW TO RECORD A SENSOR

1. Pick a segment and one of its endpoints.
2. Measure distance along that segment.
3. Enter angle around the pipe, if known.
4. Enter radius from the axis for a 3-D point.
5. Identify mount, contact or sensing junction.

Positive rotation is right-hand about the directed segment (+s). e90 = tangent x e0.

For vertical pipes, e0 is drawing +x. For other pipes, e0 is projected upward vertical.

The 45-degree dot is illustrative. No VCU sensor angle has been assumed.

A blank angle or radius leaves the exported 3-D sensor point unresolved.

J Current candidate input tables

Main-loop geometry

Segment	Endpoints	Length mm	Phi deg	Evidence
upcomer	B to C	1219.200	0.000	CAD-derived
upper_left_bend	C to D	*Unknown*	*Unknown*	CAD-derived
cooler	D to E	1352.550	100.800	CAD-derived
upper_right_bend	E to F	*Unknown*	*Unknown*	CAD-derived
downcomer	F to G	1228.669	180.000	CAD-derived
lower_right_bend	G to H	*Unknown*	*Unknown*	CAD-derived
heater	H to A	1346.496	280.794	CAD-derived
lower_left_bend	A to B	*Unknown*	*Unknown*	CAD-derived

Bends use offsets, not straight-length inputs. See the editable Geometry sheet and source length basis. The example forms do not populate these baseline cells.

Main-loop geometric sensor references

Sensor	Segment	From	Distance mm	Angle deg
STC-02	upcomer	B	116.686	*Unknown*
STC-03	upcomer	B	446.886	*Unknown*
STC-04	upcomer	C	492.914	*Unknown*
STC-05	upcomer	C	277.014	*Unknown*
LF-01	upcomer	C	61.114	*Unknown*
STC-06	cooler	D	69.142	*Unknown*
PTC-01	cooler	D	97.080	*Unknown*
STC-07	cooler	E	155.331	*Unknown*
LF-03	cooler	E	85.481	*Unknown*
PTC-03	cooler	E	85.481	*Unknown*
STC-08	downcomer	F	183.105	*Unknown*
STC-09	downcomer	F	411.705	*Unknown*
STC-10	downcomer	G	537.564	*Unknown*
STC-11	downcomer	G	283.564	*Unknown*
LF-04	downcomer	G	50.800	*Unknown*
PTC-04	heater	H	63.716	*Unknown*
STC-12	heater	H	69.293	*Unknown*
LF-02	heater	A	83.394	*Unknown*
PTC-02	heater	A	83.394	*Unknown*
STC-01	heater	A	49.428	*Unknown*

JTC-A/B are retained in Calibration, but jacket-port positions are not yet drawn. DAQ mappings, calibration and model mappings remain unknown unless separately entered with evidence.

VCU fluid choices

Prop-30, air and water are permitted candidate options from the user. A circuit/run assignment is still required. The supplied draft phase campaign is Prop-30-specific.

Fluid	Product	Circuit	Range C
PROP30	Propylene glycol / water	*Unknown*	*Unknown*
AIR	Air	*Unknown*	*Unknown*
WATER	Water	*Unknown*	*Unknown*

K Complete requirements and evidence register

The Requirements workbook tab is the editable master. Each record states why the field is needed, who should supply it, where the requirement and evidence came from, what uses it supports, and what remains unresolved. Requirement provenance is distinct from evidence for a value. **Review rule:** the records below describe candidate evidence. Approved choices require a recorded decision, scope and effective configuration; none has been established in this seed package.

C01 | Facility and configuration ID

Purpose: Tie measurements to one physical setup.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Section 3 Method, document facility configuration
Current evidence [Reported]: VCU; prototype ID VCU-REV9-DRAFT-001
Evidence source / locator: USER / current workbook; Facility selection; Start sheet
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which installed configuration and effective date does this identify?

C02 | Effective date and change history

Purpose: Explain what changed between datasets.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; DESIGN; Campaign execution conventions; proposed revision log
Current evidence [Reported]: 2026-10-16 / VCU-REV9-DRAFT-002: PTC-X removed from active sensor and calibration records at the repository owner's request. Documentation change only; no physical removal or installed-hardware verification is asserted.
Evidence source / locator: USER; Repository owner request in the Codex session, 2026-10-16; Start effective date and change summary.
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record effective date, author, changed item, reason, previous/new evidence, affected runs and whether recalibration or model revalidation is needed.

C03 | Reviewer, date and approved use

Purpose: Record an actual release decision separately from evidence type.
Package / responsible team: Shared / Facility lead / package reviewer
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Approval record and Section 3 Disposition
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Who may release the package, and for which experiment/model use?

C04 | Source revision and file hash

Purpose: Make the geometry baseline reproducible.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: R9; Sheet 8 source identity
Current evidence [CAD-derived]: Rev9 STEP SHA-256 78c6ae52bf39084f bd9b1306d28dcbe2 c1c4d5fccd334b0e 68c861022a1bfbcf
Evidence source / locator: R9; Sheet 8
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Does this source match installed VCU hardware?

C05 | Coordinate datum and axes

Purpose: Give every location the same reference.
Package / responsible team: Configuration / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: R9; Sheets 1 and 7
Current evidence [CAD-derived]: B origin; x right; y up; n normal; B-C-D-E-F-G-H-A-B
Evidence source / locator: R9; Sheets 1, 7
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Identify these points on a dated hardware photo.

C06 | Segment endpoints and straight length

Purpose: Define the length being measured.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; SEG; Method geometry map; segment rows
Current evidence [CAD-derived]: Upcomer B-C 1219.200 mm; other spans in SEG
Evidence source / locator: SEG; upcomer row: reference_length_mm and length_basis
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Measure axis-end span; do not substitute virtual-corner length.

C07 | Installed inclination and gravity direction

Purpose: Determine elevation changes and buoyancy orientation.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; R9; Method geometry map; sheets 1-2
Current evidence [CAD-derived]: Inclined CAD cylinder axes 10.800 deg to horizontal
Evidence source / locator: R9; Sheets 1-2
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: What is the measured installed tilt and uncertainty?

C08 | Bend geometry and length basis

Purpose: Avoid replacing a chord with an arc.
Package / responsible team: Configuration / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: SEG; Bend rows and length_basis
Current evidence [CAD-derived]: Three torus arcs; G-H recovered spine qualified; exact joined path unresolved
Evidence source / locator: SEG; R9; Bend rows; sheet 2
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: How will the physical bend be measured and represented?

C09 | Pipe inside diameter

Purpose: Set fluid area, inventory and hydraulic dimension.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; DIA; Geometry map; Main swept loop
Current evidence [Reported]: Main CAD bore absent; notes report nominal 1.049 in (26.64 mm)
Evidence source / locator: DIA; MAT; Main swept loop row; VCU Piping lines 23-27
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm actual bore/specification and local restrictions; keep CAD null.

C10 | Pipe outside diameter

Purpose: Set wall and external thermal geometry.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: DIA; Main swept loop row
Current evidence [CAD-derived]: 33.401 mm CAD OD
Evidence source / locator: DIA; Main swept loop / outer_diameter_mm
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm installed OD and measurement uncertainty.

C11 | Pipe material grade and certificate

Purpose: Identify the solid whose properties will be used.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; MAT; Geometry map; VCU Piping
Current evidence [Reported]: 316L, nominal 1-inch schedule 40 reported
Evidence source / locator: MAT; VCU Piping lines 23-27
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Provide purchase/specification or material certificate.

C12 | Dimension uncertainty and method

Purpose: Propagate measurement uncertainty without using CAD print precision.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; DESIGN; Measurement uncertainties; proposed geometry uncertainty field
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record method, instrument, repeatability and uncertainty units for each dimension.

C13 | Insulation product and layer order

Purpose: Identify the actual thermal stack.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; MAT; Geometry map; VCU insulation
Current evidence [Reported]: Main: fiberglass 5556K73; heater: calcium aluminum silicate 4057K14 reported
Evidence source / locator: MAT; Lines 29-44
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which product and layers are installed on this exact segment?

C14 | Insulation thickness by section

Purpose: Set the thermal resistance geometry.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; MAT; Geometry map; VCU insulation
Current evidence [Reported]: Most piping about 2 in; heater insulation 1/4 in reported
Evidence source / locator: MAT; Lines 29-44
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Measure each layer; do not apply either value everywhere.

C15 | Coverage start/end, gaps and condition

Purpose: Capture uninsulated heat-loss paths.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Method: facility and fluid configuration; geometry map
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

C16 | Insulation contact and support details

Purpose: Represent contact resistance and support heat paths.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; HEAT; Geometry map; VCU Insulation
Current evidence [Modeling assumption]: Firm contact is explicitly an assumption in notes
Evidence source / locator: HEAT; VCU Insulation lines 7-10
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Is there evidence of contact, compression, gaps or support conduction?

C17 | Dated overview and detail photos

Purpose: Make the configuration recognizable and auditable.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: USER; DESIGN; Requested form-like geometry package
Current evidence [Reported]: Supplied images are illustrations; not a dated installed survey
Evidence source / locator: USER; Supplied VCU image captions
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Attach photos showing endpoints, insulation and sensors with scale.

C18 | Fluid identity and circuit assignment

Purpose: Avoid confusing working fluid, coolant and room air.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: USER; P0; Fluid scope; facility/fluid configuration
Current evidence [Reported]: VCU options: Prop-30, air, water; no circuit assignment established
Evidence source / locator: USER; Fluid clarification in chat
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which fluid occupies each circuit in this run?

C19 | Composition and concentration basis

Purpose: Select the correct property family.
Package / responsible team: Configuration / Experimental team
Applies to: VCU mixtures
Requirement source / locator: P0; Method fluid configuration
Current evidence [Reported]: Prop-30: 30% propylene glycol by mass reported
Evidence source / locator: MAT; PG; MAT lines 19-21; PG 30 wt% note
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Identify formulation, preparation and concentration verification.

C20 | Batch, fill inventory and condition

Purpose: Reproduce thermal capacity and hydraulic state.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; P2; P0 fluid configuration; P2 fill record
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record batch, fill date, mass/level, contamination and trapped-gas evidence.

C21 | Sensor hardware ID and type

Purpose: Identify the installed instrument independently of a CAD label.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; SNS; Instrumentation documentation; sensor rows
Current evidence [CAD-derived]: 12 STC, 4 LF, 5 loop PTC and 2 JTC geometric references
Evidence source / locator: R9; Sheet 5 inventory
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Verify installed hardware count and channel identities for remaining sensors. PTC-X was removed from the active register at the repository owner's request on 2026-10-16; no physical removal is asserted.

C22 | Sensor model and serial number

Purpose: Connect the device to its calibration evidence.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method instrumentation documentation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

C23 | DAQ channel crosswalk

Purpose: Connect a physical sensor to the recorded signal.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method instrumentation documentation
Current evidence [Unknown]: Unverified; STEP does not encode acquisition-channel labels
Evidence source / locator: SNS; channel_label_in_CAD blank; label_basis
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Trace cable/DAQ configuration and verify the mapping with a recorded check.

C24 | Sensor segment, endpoint and axial distance

Purpose: Locate the sensor without scaling an image.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: SNS; nearest_endpoint, nearest_distance_mm
Current evidence [CAD-derived]: STC-02: upcomer, B + 116.686 mm geometric reference
Evidence source / locator: SNS; sensor_id=STC-02
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Measure the actual sensing/contact reference and identify the point basis.

C25 | Circumferential angle and zero reference

Purpose: Locate a sensor around the pipe circumference.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: USER; DESIGN; Circumference clarification; coordinate guide
Current evidence [Modeling assumption]: Convention proposed: projected top; vertical pipe uses drawing +x
Evidence source / locator: DESIGN; Coordinate convention
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm convention; measure angle. No sensor angles have been supplied.

C26 | Radius, insertion depth and sensing extent

Purpose: Distinguish mount, sheath, junction and fiber active region.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; SNS; Installed coupling; reference_basis
Current evidence [CAD-derived]: CAD does not define verified TC junctions or optical sensing extents
Evidence source / locator: SNS; R9; reference_basis; sheets 5-6
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record physical junction or fiber extent; a mount centroid is not a tip.

C27 | Attachment and thermal coupling

Purpose: Interpret what temperature a sensor measures.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Method instrumentation documentation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Describe attachment, contact, insulation position and time response.

C28 | Calibration date, certificate and validity

Purpose: Establish calibration provenance.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; DESIGN; Method instrumentation documentation; proposed maintenance/validity extension
Current evidence [Reported]: Plan mentions a prior five-temperature bath; date/certificate not supplied
Evidence source / locator: P0; Section 2 apparatus, channel table
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Provide instrument-specific date, certificate, traceability, next-due or recheck trigger, and validity at the run date.

C29 | Calibration range and correction

Purpose: Apply corrections only where supported.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method instrumentation documentation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply calibration points, applied correction and residuals with units.

C30 | Measurement uncertainty and coverage

Purpose: Combine uncertainties on a consistent basis.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P1; Instrumentation documentation; Data Acceptance and Uncertainty
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: State standard/expanded uncertainty, coverage factor, confidence and shared error sources.

C31 | Sampling, scaling, filtering and response

Purpose: Reproduce recorded signals and temporal response.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Instrumentation documentation and logging
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

C32 | Clock synchronization and common-event check

Purpose: Align measurements, commands and events.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method logging and time synchronization
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

C33 | Mandatory / alternate / diagnostic role

Purpose: Define usable channels before selecting results.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P0; P1; P0 acceptance-channel matrix; P1 TC-Luna comparison
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which channels are mandatory, and which alternates are approved with evidence?

C34 | Heater element identity and active spans

Purpose: Map where electrical heat is generated.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; HEAT; R9; Geometry map; heater note; sheet 4
Current evidence [Reported]: Four clamps in CAD; notes report mat/band electrical heaters and intended 40-in coverage
Evidence source / locator: HEAT; R9; HEAT lines 16-23; R9 sheet 4
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Survey each energized element; clamp edges and reported total coverage are not active boundaries.

C35 | Heater supply and circuit diagram

Purpose: Establish the electrical measurement boundary.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: USER; HEAT; Requested power/circuit schematic; heater note
Current evidence [Reported]: Shared voltage source is described as in theory
Evidence source / locator: HEAT; Line 19
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Provide rated elements, wiring, series/parallel layout, switching and measurement points.

C36 | Measured heater power and uncertainty

Purpose: Distinguish input power, controller command and heat entering the loop.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P3; DESIGN; Secondary duty; electrical-heater adaptation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Provide time-resolved real electrical power, meter/waveform basis, losses and storage accounting.

C37 | Heater control and monitor signals

Purpose: Explain controller actions and protect measurement interpretation.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method logging; approved operating limits
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record feedback sensor, setpoint, output, on/off state, alarms and wiring reference.

C38 | Cooler jacket and active cooling boundaries

Purpose: Separate mechanical geometry from effective heat transfer.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: R9; Sheet 3
Current evidence [CAD-derived]: Shell 1017.5875 mm; complete jacket 1030.2875 mm
Evidence source / locator: R9; Sheet 3; J-IN/J-OUT
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Measure active span and preserve shell versus complete-envelope definitions.

C39 | Coolant routing and measurement planes

Purpose: Bound the measured cooler energy balance.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Method secondary-side energy-balance boundaries
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm actual inlet/outlet, intervening hoses, bypasses, heat leaks and sensor planes.

C40 | Coolant flow and inlet/outlet measurements

Purpose: Compute duty with calibrated signals and correct coolant properties.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Secondary-side boundaries
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply flow-meter basis, calibration, fluid properties, temperatures and signs.

C41 | Cooler feedback and setpoint history

Purpose: Distinguish an imposed model condition from actual control behavior.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method logging and time synchronization
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

C42 | TLEP/BREP connections and termination

Purpose: Locate branch interfaces and attached hardware.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: R9; Sheet 4
Current evidence [CAD-derived]: Each modeled exit tube is 469.900 mm; IN/OUT means near/far, not flow direction
Evidence source / locator: R9; Sheet 4 endpoints and connection qualifications
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm valve/tank connectivity, actual cut plane and installed branch geometry.

C43 | Tank level, vent/pressure and valve states

Purpose: Define hydraulic and thermal branch conditions.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method document facility/fluid configuration
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record fill level, gas/vent pressure, valve/bypass state and changes during the run.

C44 | Ambient temperature/RH and outer-surface locations

Purpose: Resolve room gradients and condensation conditions.
Package / responsible team: Configuration / Experimental team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P0; Method environmental and condensation measurements
Current evidence [Reported]: Draft calls for two room T/RH stations and cold exposed-surface measurements
Evidence source / locator: P0; Method environmental measurements
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Are these channels installed? Under-insulation wall TCs are not exposed-surface measurements.

C45 | Operating limits and alarm/interlock basis

Purpose: Reference facility-approved limits for this setup.
Package / responsible team: Configuration / Facility lead / package reviewer
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; Method approved operating limits
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Provide approved limits and source; do not infer them from property-fit ranges.

C46 | Run ID, raw files and event log

Purpose: Tie data, changes and analysis intervals to the configuration.
Package / responsible team: Configuration / Experimental team
Applies to: VCU; every configuration and affected run
Requirement source / locator: P0; P4; Campaign execution conventions; Phase 4 records
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Supply evidence and request review.

M01 | Intended use and prediction targets

Purpose: Define what comparison would be useful.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: USER; MODEL; Both models; four model stages
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Identify outputs, required accuracy and excluded uses for each model.

M02 | Model/software/input revision

Purpose: Reproduce an analysis with the same inputs.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: MODEL; P2; Run outputs; model freeze
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record solver/code revision, resolved configuration and property version.

M03 | Model domain and geometry simplifications

Purpose: Make excluded branches and healed geometry visible.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: R9; MODEL; Sheet 8 qualifications; model stages
Current evidence [CAD-derived]: Heater overlap/offset and G-H path require an explicit interpretation
Evidence source / locator: R9; Sheets 2, 8
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which simplifications are accepted and what sensitivity supports them?

M04 | 1-D cell and CFD region mapping

Purpose: Compare the right model quantity with each sensor.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Method geometry/model map
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Map physical sensing points or averages to cells/regions; include non-coincident offsets.

M05 | Flow mode and governing equations

Purpose: Identify zero-flow, prescribed-flow or buoyancy-coupled assumptions.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: MODEL; The four physical-model stages
Current evidence [Reported]: M0-M3 and a separate reduced-zone model are described
Evidence source / locator: MODEL; Stage table and reduced-zone paragraph
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Select governing equations, gravity/pressure reference, compressibility or Boussinesq approximation, and laminar/turbulent treatment with validity; identify any effective-zone assumptions.

M06 | Storage, conjugate heat transfer and contact model

Purpose: Represent fluid, wall, insulation and interface physics.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: MODEL; P2; Model stages; reduced-zone thermal model
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Which layers, axial paths, capacitances and contacts are included?

M07 | TLEP/BREP mathematical boundary conditions

Purpose: Translate documented branch hardware into equations.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: USER; P0; Exit-pipe alignment request; configuration record
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Specify termination plane, mass/pressure and thermal conditions; no throughflow does not imply adiabatic.

M08 | Heater source distribution and power balance

Purpose: Refer electrical measurements to the modeled control volume.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P3; DESIGN; Duty definitions; VCU electrical adaptation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Specify spatial source, meter boundary, external losses and storage without double counting.

M09 | Cooler mathematical boundary and feedback

Purpose: Match model inputs to measured coolant/control conditions.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; MODEL; Secondary boundaries; cooler forms
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Choose measured inlet/flow, imposed wall temperature or coefficient model with rationale.

M10 | External convection and radiation model

Purpose: Represent exchange with room and surfaces.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P2; MODEL; Reduced-zone heat loss; thermal network
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record ambient assignment, emissivity, surface state, correlations and validity.

M11 | Hydraulic friction and local losses

Purpose: Compute resistance and uncertainty consistently.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: MODEL; DESIGN; M3 momentum model; proposed closure record
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record roughness, K values, correlation versions, reference diameters and regime limits.

M12 | Material/product-to-property link

Purpose: Use properties for the actual material or fluid.
Package / responsible team: Shared / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Definitions: fixed property reference
Current evidence [Reported]: Reported 316L and insulation products; exact fluid formulation unresolved
Evidence source / locator: MAT; VCU section lines 17-44
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Confirm installed identity; list needed fluid density/cp/enthalpy/viscosity/k/expansion and solid density/cp/k by model, plus relevant surface/contact properties.

M13 | Property source and exact equation/table

Purpose: Make every numerical property reproducible.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Definitions and Conventions
Current evidence [Reported]: Insulation k(T) candidate exists in notes; no correlation approved
Evidence source / locator: MAT; STEEL; PG; MAT lines 35-40; STEEL Suggested you use; PG notes
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Retain original publication/product table and exact coefficients, edition and locator.

M14 | Input/output units and concentration basis

Purpose: Prevent temperature and mixture-basis conversion errors.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; Definitions: property reference
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record K versus deg C, pressure units, mass/volume fraction and output units for each function.

M15 | Applicability domain

Purpose: Prevent unsupported extrapolation across fluids or regimes.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P2; Property reference; temperature/condensation applicability
Current evidence [Reported]: Insulation fit is reported valid 297-533 K; not an experiment operating range
Evidence source / locator: MAT; Lines 38-40
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Verify property-specific T/p/composition/phase domain from original sources.

M16 | Interpolation and out-of-range behavior

Purpose: Make tables and correlations behave consistently.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: DESIGN; Proposed implementation control
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Specify interpolation; reject/flag extrapolation explicitly rather than silently clipping.

M17 | Property uncertainty and fit residuals

Purpose: Separate source uncertainty from fitting accuracy.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P1; Property reference; uncertainty propagation
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Quantify source, composition, fit and implementation uncertainty separately.

M18 | Selection rationale and alternatives

Purpose: Explain why both teams use this property model.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: USER; Requested material-property justification
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Compare candidate sources and explain selection or any deliberate mixed-source set.

M19 | Implementation reference checks

Purpose: Check equation transcription and thermodynamic consistency.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: DESIGN; Proposed shared property tests
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Store reference values; check cp/enthalpy and density/expansion consistency where used.

M20 | Initial temperature, flow and inventory state

Purpose: Initialize transient models from evidence.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P2; Method initial condition; Required Outputs
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record fluid/wall distribution, initial flow, fill history and unresolved state uncertainty.

M21 | Observation averaging and sensor response

Purpose: Compare like quantities rather than convenient nodes.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P0; P2; Model map; effective initial states
Current evidence [Reported]: Reduced-zone temperature is an effective state, not automatically fluid/wall temperature
Evidence source / locator: P2; MODEL; Reduced-zone model; reduced-zone paragraph
Review state: Needs review
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Define spatial/time averaging, lag and fluid-wall weighting.

M22 | Steady-state class and acceptance limits

Purpose: Fix data-quality criteria before selecting windows.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P1; Section 2 Data Acceptance and Uncertainty
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Approve intended-use limits and demonstrate resolvability; do not loosen them to fit noise.

M23 | Accepted interval and target-attainment result

Purpose: Keep constancy separate from reaching the requested target.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P4; Method establishing/recording a state
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Record acceptance time, confirmed span, reporting interval and target-band result separately.

M24 | Covariance and autocorrelation method

Purpose: Report uncertainty using independent information.
Package / responsible team: Modeling / Computational team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P1; Required Outputs: correlation structure and uncertainty
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Freeze block/window rules, Type A/B and shared-error covariance with evidence.

M25 | Calibration/withheld split and model freeze

Purpose: Keep validation independent of parameter fitting.
Package / responsible team: Modeling / Computational team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P2; P3; P4; P2 runs/freeze; P3 blind validation; P4 applicability case
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Declare run roles before fitting; record model/property/mapping versions and any reassignment.

M26 | Parameter identifiability and sensitivities

Purpose: Avoid fitting unsupported model complexity.
Package / responsible team: Modeling / Computational team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P2; Required Outputs steps 5-7
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Report parameter covariance, residual structure, start-time and initial-state sensitivity.

M27 | Settling time and supported domain

Purpose: Use a supported operational settling value.
Package / responsible team: Modeling / Computational team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P2; Slowest loop time constant; Required Outputs
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Report uncertainty and conservative supported bound; do not use an archived plateau fit.

M28 | Mass/energy conservation and solution convergence

Purpose: Check that the numerical solution is credible.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: MODEL; DESIGN; Heat ledger; proposed verification record
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Define residual/closure criteria and failure handling for each solver.

M29 | Spatial/time resolution sensitivity

Purpose: Separate discretization error from experimental discrepancy.
Package / responsible team: Modeling / Computational team
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: DESIGN; Proposed 1-D and CFD verification field
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Report mesh/cell/time refinement and tolerance sensitivity relevant to predictions.

M30 | Flow inference and conditioning gates

Purpose: Avoid reporting flow from unresolved enthalpy differences.
Package / responsible team: Modeling / Computational team
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P4; Section 2 Primary-flow and Reynolds-number inference
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Use validated loss corrections, covariance-aware consistency and defined uncertainty gates.

M31 | Validation metrics, discrepancy and applicability

Purpose: Bound what has actually been demonstrated.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU; 1-D and CFD / CHT
Requirement source / locator: P2; P3; P4; Validation and disposition sections
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Define metrics and acceptance basis; preserve blind comparisons and model limitations.

M32 | Repeatability and path-dependence criteria

Purpose: Distinguish significant history effects from measurement scatter.
Package / responsible team: Modeling / Experimental + computational
Applies to: VCU Prop-30 draft campaign; requalify other fluids
Requirement source / locator: P4; Method central-state repeat; Required Outputs
Current evidence [Unknown]: Unknown
Evidence source / locator: None supplied; Not available
Review state: Needs information
Selected choice: None selected
Decision scope / rationale: *Unknown*
Reviewer / date; effective configuration: Pending; Unknown
To resolve: Preapprove response set, equivalence limits and practical-significance threshold.

L Current model and material input records

These records reproduce the existing Model choices and Materials worksheets. Unknown inputs remain visible. New answers in the three prototype forms are evidence for review; they do not automatically populate these records.

L.1 Model choices

TLEP and expansion tank

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Confirm level, vent/pressure, connectivity and thermal coupling.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

BREP and drain

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Confirm valve state, termination plane and thermal coupling.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Heater power

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Electrical heater evidence conflicts with draft secondary-heater plan.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Cooler control

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Document coolant flow, inlet temperature, feedback and measurement planes.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

External convection / radiation

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Document room environment, coverings, surfaces and supports.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Initial conditions

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Fluid, wall, insulation, inventory and initial flow.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Flow / heat-transfer closures

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Record correlation, regime, reference dimensions and domain.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Verification and validation

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: Conservation, mesh/time sensitivity, calibration/withheld split and metrics.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

Plan approval

Observed state: *Unknown*
1-D choice: *Unknown*
CFD / CHT choice: *Unknown*
Evidence / rationale: STNCL plans are drafts until approval status is confirmed.
Uncertainty / sensitivity: *Unknown*
Owner: *Unknown*
Status: Unknown

L.2 Materials

PIPE

Identity / product: 316L stainless steel
Property: rho, cp, k
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: Reconcile candidate correlations and installed specification.
Status: Reported
Source: 1.2 Main Material Props.md; 1.2.2 Steel Pipe Properties.md

MAIN-INS

Identity / product: Fiberglass / reported McMaster 5556K73
Property: rho, cp, k
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: Confirm installed layers and original product evidence.
Status: Reported
Source: 1.2 Main Material Props.md

HEATER-INS

Identity / product: Calcium aluminum silicate / reported 4057K14
Property: rho, cp, k
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: Confirm installed stack and contacts.
Status: Reported
Source: 1.2 Main Material Props.md

PROP30

Identity / product: Product/formulation unresolved
Property: rho, cp, h, mu, k, beta
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: 30% by mass reported; identify original property source.
Status: Reported
Source: 1.2.1 Prop 30 ThermoPhys Props.md

AIR

Identity / product: Air
Property: rho, cp, h, mu, k
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: Define humidity, pressure and temperature domain.
Status: Unknown
Source: User fluid scope

WATER

Identity / product: Water
Property: rho, cp, h, mu, k, beta
Equation / table: *Unknown*
Input units: *Unknown*
Output units: *Unknown*
Valid domain: *Unknown*
Uncertainty: *Unknown*
Rationale: Define purity, pressure and temperature domain.
Status: Unknown
Source: User fluid scope

L.3 Calibration summary

Complete certificates, corrections and acquisition details remain on Calibration and in the resolved input export.

Sensor	Last calibration	Uncertainty	Model mapping
STC-02	*Unknown*	*Unknown*	*Unknown*
STC-03	*Unknown*	*Unknown*	*Unknown*
STC-04	*Unknown*	*Unknown*	*Unknown*
STC-05	*Unknown*	*Unknown*	*Unknown*
LF-01	*Unknown*	*Unknown*	*Unknown*
STC-06	*Unknown*	*Unknown*	*Unknown*
PTC-01	*Unknown*	*Unknown*	*Unknown*
STC-07	*Unknown*	*Unknown*	*Unknown*
LF-03	*Unknown*	*Unknown*	*Unknown*
PTC-03	*Unknown*	*Unknown*	*Unknown*
STC-08	*Unknown*	*Unknown*	*Unknown*
STC-09	*Unknown*	*Unknown*	*Unknown*
STC-10	*Unknown*	*Unknown*	*Unknown*
STC-11	*Unknown*	*Unknown*	*Unknown*
LF-04	*Unknown*	*Unknown*	*Unknown*
PTC-04	*Unknown*	*Unknown*	*Unknown*
STC-12	*Unknown*	*Unknown*	*Unknown*
LF-02	*Unknown*	*Unknown*	*Unknown*
PTC-02	*Unknown*	*Unknown*	*Unknown*
STC-01	*Unknown*	*Unknown*	*Unknown*
JTC-A	*Unknown*	*Unknown*	*Unknown*
JTC-B	*Unknown*	*Unknown*	*Unknown*

M Source catalog and limits of use

Use the source ID with the row/page/section locator. Source files are not operating instructions. Exact local paths and read-time file hashes are recorded in support/reference/source_manifest.json. Files remain at their supplied locations; they are not all duplicated into this repository.

R9 | VCU_Rev9_CAD_Engineering_Drawings.pdf

Revision / context: 2026-10-08; VCU-R9-E01
Find it: Sheets 1-8
Qualification: VCU CAD candidate. Printed precision is not tolerance; geometry is not installed verification.

SEG | VCU_Rev9_Segments.csv

Revision / context: Companion Rev9 structured package
Find it: Rows keyed by segment; length_basis, dx_mm, dy_mm, centerline_length_mm
Qualification: Preserve blanks, projected spans and qualified spline lengths.

SNS | VCU_Rev9_Sensor_Stations.csv

Revision / context: Companion Rev9 structured package
Find it: Rows keyed by sensor_id; reference_basis, nearest_endpoint, nearest_distance_mm
Qualification: Geometric references, not verified sensing tips or DAQ names.

DIA | VCU_Rev9_Diameters.csv

Revision / context: Companion Rev9 structured package
Find it: Rows keyed by item
Qualification: Main swept-loop bore is absent; local spool bores are not loop-wide IDs.

Source catalog — continued

MAT | 1.2 Main Material Props.md

Revision / context: Supplied notes; VCU section cites 2026-09-16 report
Find it: VCU section only; lines 17-44
Qualification: Reported materials/composition and proposed insulation fit; original sources not verified.

PG | 1.2.1 Prop 30 ThermoPhys Props.md

Revision / context: Supplied notes
Find it: 30 wt% heading and linked property figures
Qualification: Property-source leads; do not transfer image/table values without source and units.

STEEL | 1.2.2 Steel Pipe Properties.md

Revision / context: Supplied notes
Find it: Lines 3-21; Suggested you use
Qualification: Mixed candidate correlations. A range beside MOOSE items does not qualify the whole selection.

HEAT | 1.1.0 Heaters and coolers.md

Revision / context: Supplied notes; 2026-10-07 reported heater message
Find it: VCU / Insulation / Heater; lines 5-23
Qualification: Electrical mat/band heaters reported. Wiring, active boundaries and net duty unverified.

Source catalog — continued

SNOTE | 1.1.2 Sensors.md

Revision / context: Supplied notes
Find it: VCU / Rev9; PTCX note
Qualification: PTCX called a fluke; some links in this section point to ACU/Rev8. Keep conflict visible.

MODEL | 0.0 Intro to Model.md

Revision / context: Supplied model overview
Find it: The four physical-model stages; reduced-zone paragraph
Qualification: Supports proposed model scope; not verification that any facility run is validated.

P0 | STNCL_Phase0_System_Checkout.tex

Revision / context: Revision 0.1; 2026-08-17; draft
Find it: Section 2 Technical Basis; Section 3 Method / Required Outputs
Qualification: Draft requirements for facility-specific completion; approval not supplied.

P1 | STNCL_Phase1_Ambient_Equilibrium_Baseline.tex

Revision / context: Revision 0.1; draft
Find it: Section 2 Data Acceptance and Uncertainty; Section 3 Method / Required Outputs
Qualification: Prop-30 plan; thresholds and noise examples are not universal operating limits.

Source catalog — continued

P2 | STNCL_Phase2_Thermal_Relaxation_Tests.tex

Revision / context: Revision 0.1; draft
Find it: Section 2 Thermal Model and Heat Loss; Section 3 Method / Required Outputs
Qualification: Prop-30 plan; conditioning and tank qualification wording need reconciliation.

P3 | STNCL_Phase3_Low_Duty_Steady_Heat_Loss_Tests.tex

Revision / context: Revision 0.1; draft
Find it: Section 2 Blind validation; Section 3 Method / Required Outputs / Disposition
Qualification: Heater-side duty definition requires electrical adaptation for VCU.

P4 | STNCL_Phase4_Initial_Test_Matrix_and_Transition_Characterization.tex

Revision / context: Revision 0.1; draft
Find it: Section 2 Primary-flow inference; Section 3 Method / Required Outputs
Qualification: Five unique states plus repeat; applicability and uncertainty gates are draft requirements.

USER | User decisions in this chat

Revision / context: 2026-10-15
Find it: Editable forms; VCU fluid scope; both models; circumferential sensor position
Qualification: VCU Prop-30/air/water; circuit and run assignment still required. ACU excluded.

Source catalog — continued

DESIGN | Proposed package design

Revision / context: 2026-10-15
Find it: This requirements register and coordinate convention
Qualification: Assistant-proposed documentation field/convention, not an approved physical fact.

Context retained outside the VCU baseline

ACU general-arrangement drawings, ANEC-000126 sensor labels, ACU sensor naming notes and ACU images belong to ACU. They supply no VCU values or channel mappings here. The supplied VCU overview PNGs identify Rev8 as their source and are visual-format references only. Mixed geometry archives and note links require member-level facility/revision checks; their contents are not imported wholesale.

Generated record identity

Workbook SHA-256 (spaces only for wrapping): 146dbd5127c4c68d e94917b5e921df48 36e4b7dd2c24c7e5 32c70d40f2c32322

All form answers, source rows and requirement fields are exported to support/generated/resolved_inputs.json. The readable register is also available in support/generated/requirements_and_evidence.md. A successful build verifies the input structure and drawing geometry, not experimental or modeling readiness.